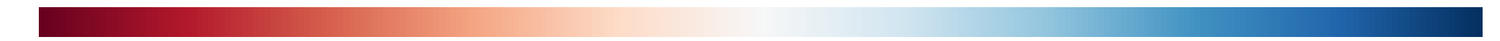
Helion speedup for every GB300 benchmark case

Combined forward records · K3 updated on GPU 3 · other 13 kernels historical · 168/168 recorded comparisons passed

	Triton request						CuTe request					
	Case 0	Case 1	Case 2	Case 3	Case 4	Case 5	Case 0	Case 1	Case 2	Case 3	Case 4	Case 5
K1 Grouped BF16 GEMM · QuACK	0.697×	0.783×	0.730×	0.853×	0.887×	0.839×	0.763×	0.884×	0.891×	1.020×	1.013×	1.025×
K2 Scaled FP8 GEMM · vLLM	0.966×	0.935×	0.855×	0.761×	0.708×	0.671×	1.000×	0.996×	1.080×	1.079×	0.881×	1.101×
K3 MXFP8 dense GEMM · DeepGEMM	0.646×	0.331×	0.611×	0.199×	0.280×	0.471×	1.108×	1.082×	1.081×	1.045×	1.071×	1.028×
K4 Variable-length top-k · FlashInfer	0.707×	0.407×	0.502×	0.680×	0.346×	0.529×	1.274×	1.200×	1.012×	1.089×	1.224×	1.005×
K5 MoE softmax top-k · SGLang	1.462×	0.969×	0.947×	1.263×	1.242×	1.317×	1.571×	1.253×	1.394×	1.527×	1.507×	2.295×
K6 Top-k / top-p sampling · FlashInfer	0.881×	1.313×	1.305×	4.048×	1.470×	0.842×	0.813×†	1.227×†	1.304×†	3.788×	1.375×†	1.193×
K9 RMSNorm · FlashInfer	1.074×	1.003×	0.914×	0.998×	1.008×	1.104×	1.281×	1.032×	1.328×	0.996×	1.059×	1.587×
K10 Fused add + RMSNorm · QuACK	0.795×	0.984×	1.003×	0.762×	0.956×	1.011×	1.031×	1.386×	1.046×	1.053×	1.335×	1.011×
K15 Cross entropy · QuACK	0.848×	0.828×	0.714×	0.888×	0.853×	1.089×	0.970×	1.188×	1.020×	1.189×	1.047×	1.200×
K16 GRPO loss · FLA	1.095×	0.925×	0.769×	0.996×	0.880×	0.707×	1.223×	1.022×	0.998×	1.130×	1.031×	0.999×
K17 NVFP4 quantization · Transformer Engine	1.462×	1.305×	0.863×	1.800×	1.266×	0.875×	1.481×	1.240×	0.997×	1.784×	1.189×	1.000×
K18 MXFP8 quantization · FlashInfer	1.096×	1.091×	1.032×	1.107×	1.118×	1.127×	1.096×	1.071×	1.032×	1.297×	1.118×	1.129×
K19 Prerouted BF16 MoE · SGLang	1.825×	1.825×	1.825×	1.841×	1.841×	1.840×	1.798×	1.799×	1.798×	2.085×	2.085×	2.085×
K20 MoE token permutation · vLLM	2.893×	2.664×	2.870×	2.657×	2.862×	2.677×	3.321×	2.975×	3.330×	2.975×	3.321×	2.967×



Baseline latency / Helion latency · log₂ color scale · white = 1×

† Includes Triton stages or a fallback. K3 is CuTe throughout; case 1 is M8. All K16 cases retain β=0.1.

Original _steady_bench · CUDA graphs · five runs · cold L2 · 1400 W · Helion bfc25902 + CuTe patches (K3, K4, K6, K19) · K3 backend: 18a14109 (11 files)